Alibaba's HanGuang 800 NPU: An Inference Processor

Sparsh Mittal

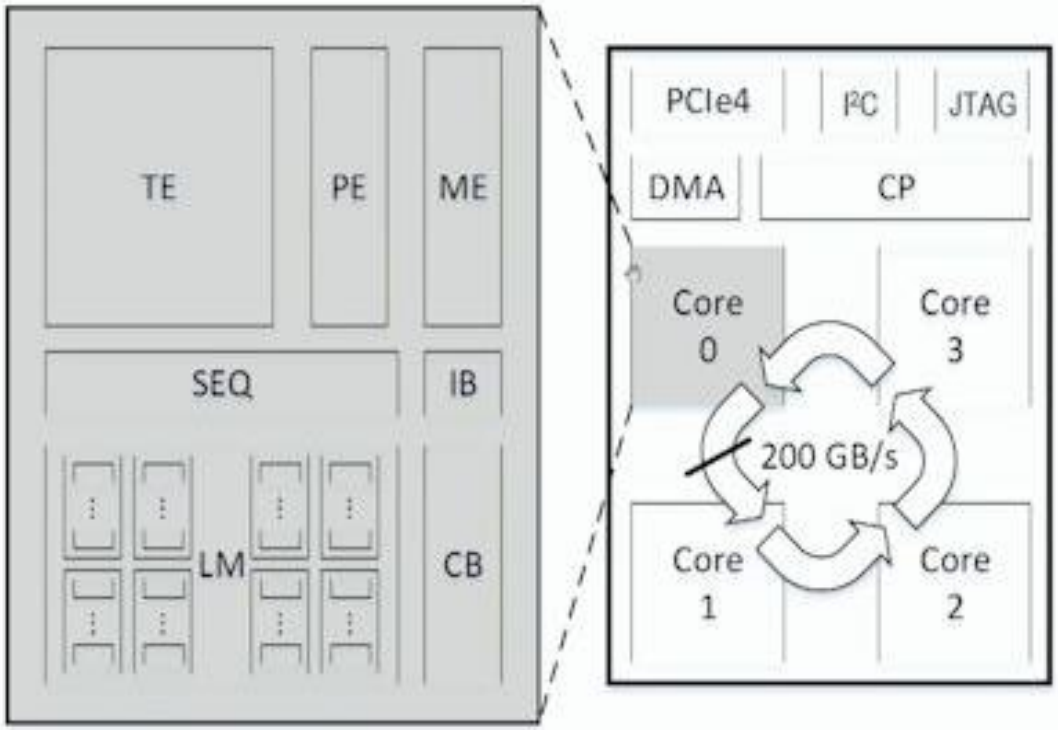
Comparative results

	Hanguang 800	Habana Goya	Cambricon MLU270	NVIDIA T4	NVIDIA P4
Fab Process	12 nm	16 nm	16 nm	12 nm	16 nm
Transistors	17 billion	NA	NA	13.6 billion	7.2 billion
Performance (IPS)	78,563	15,433	10,000	5,402	1,721
Peak Efficiency (IPS/W)	500	150	143	78	52

Performance and peak efficiency are on ResNet-50
IPS = images per second

Top Level Diagram

- Top Level:
  - 4 Cores w/ a ring bus
  - Command Processor (CP)
  - PCIE gen4 X16
- Each Core:
  - Tensor Engine (TE)
  - Pooling Engine (PE)
  - Memory Engine (ME)
- SRAM-Only:
  - 192MB Local Memory (LM)
  - Distributed shared
  - No DDR



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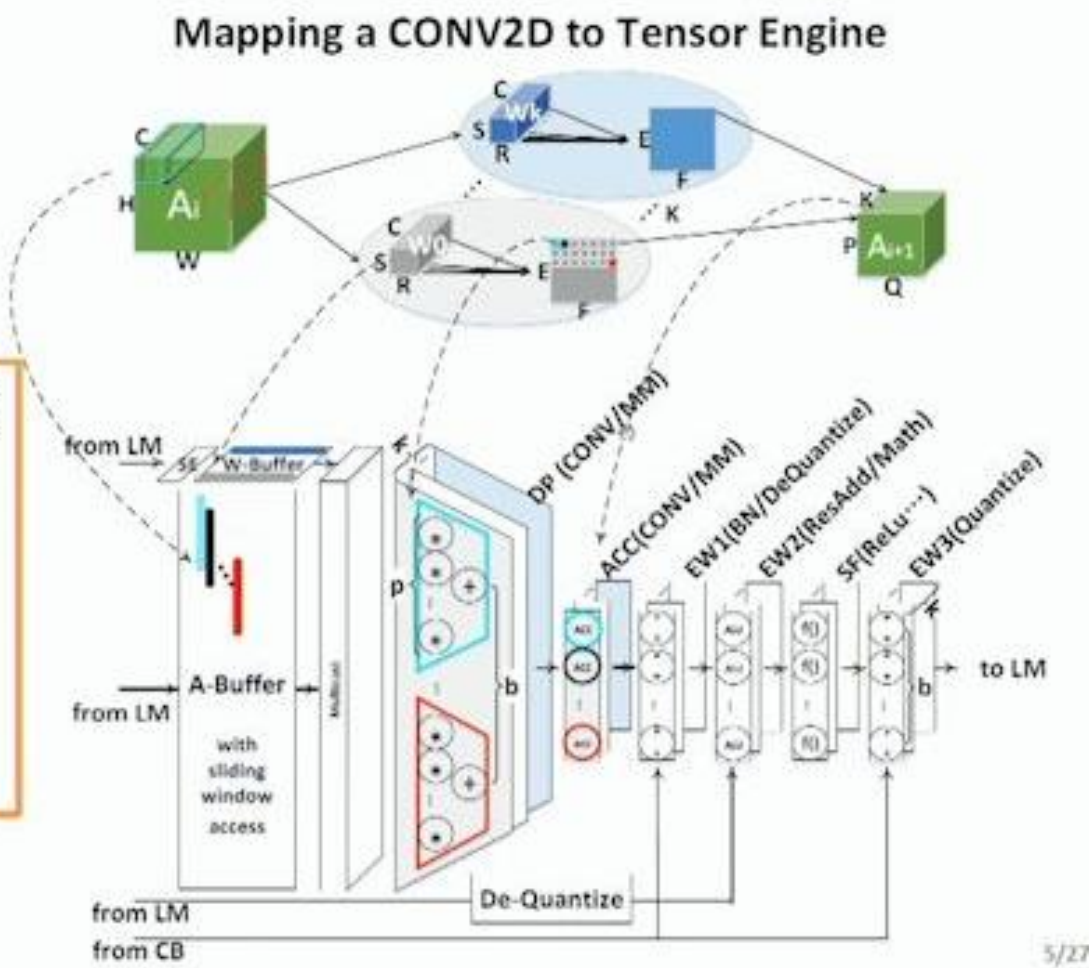
CP = control processor. DDR means DDR DRAM

Tensor Engine

- Weight+Activation Stationary
  - Data reuse
    - via W-buffer, A-buffer, Multicasting
  - Avoid img2col()
    - via sliding-window access
- Fused OPs
  - e.g. CONV-BN-ResidualADD-ReLU



EW = element-wise

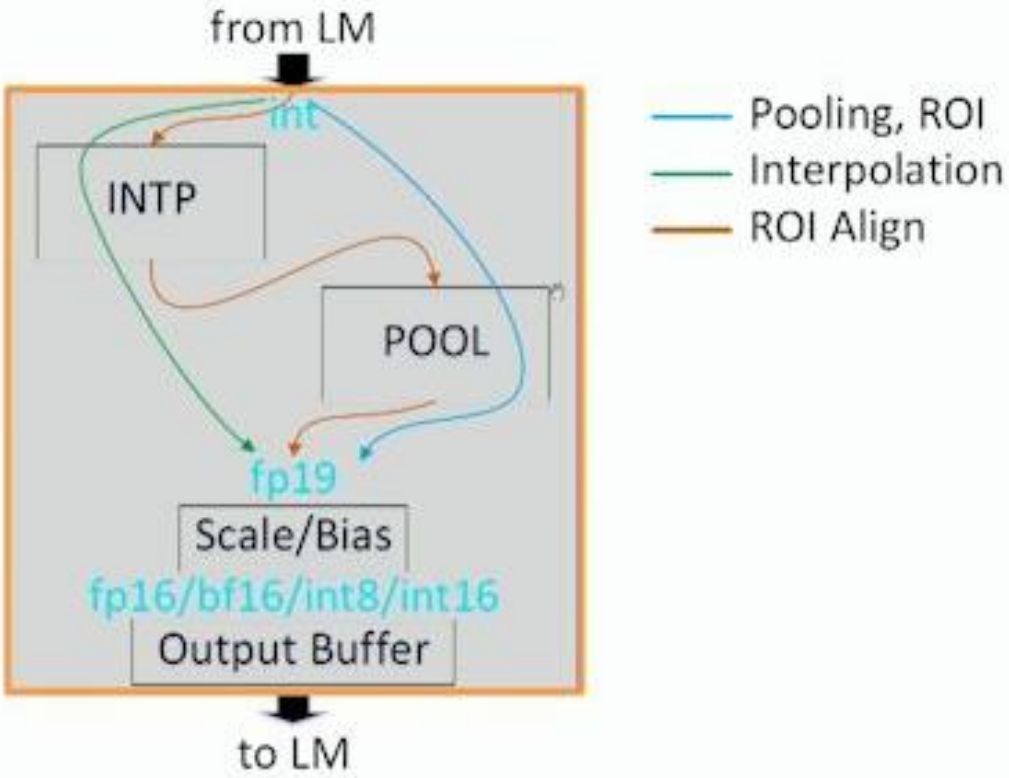
- The design has only four cores and can divide models among them, delivering peak throughput at any batch size.
- ==> HanGuang is suitable for real-time workloads that require minimum latency.
- It uses dot-product units and not systolic arrays.
  - This increases MAC utilization reduces memory requirements.
  - This, in turn, allows the chip to operate entirely without external DRAM, packing neural networks into its 192MB of SRAM.
- HanGuang achieves higher MAC utilization compared to GPUs

- Accelerated convolution and matrix multiplication with support for inverse convolution, hole convolution, 3D convolution, interpolation, and ROI.
- In addition to INT8/INT16 quantization acceleration, it also covers FP16/BFP16 vector calculations that directly accelerate various ReLu, Sigmoid, Tanh, etc.

- Domain specific CISC-like instruction set
- To support workloads larger than 192 MB, need to use multiple chips and perform chip-to-chip communication through PCIe
- Sparsity engine supports sparsity of weights only, not activations
  - This further optimizes memory usage.

Pooling Engine

- POOL Unit
  - POOLs, ROIs
  - ROI_align_2
- INTP Unit
  - Interpolations
  - ROI_align_1
- Scale/Bias Unit
  - Scaling/Bias operations
  - Data formatting



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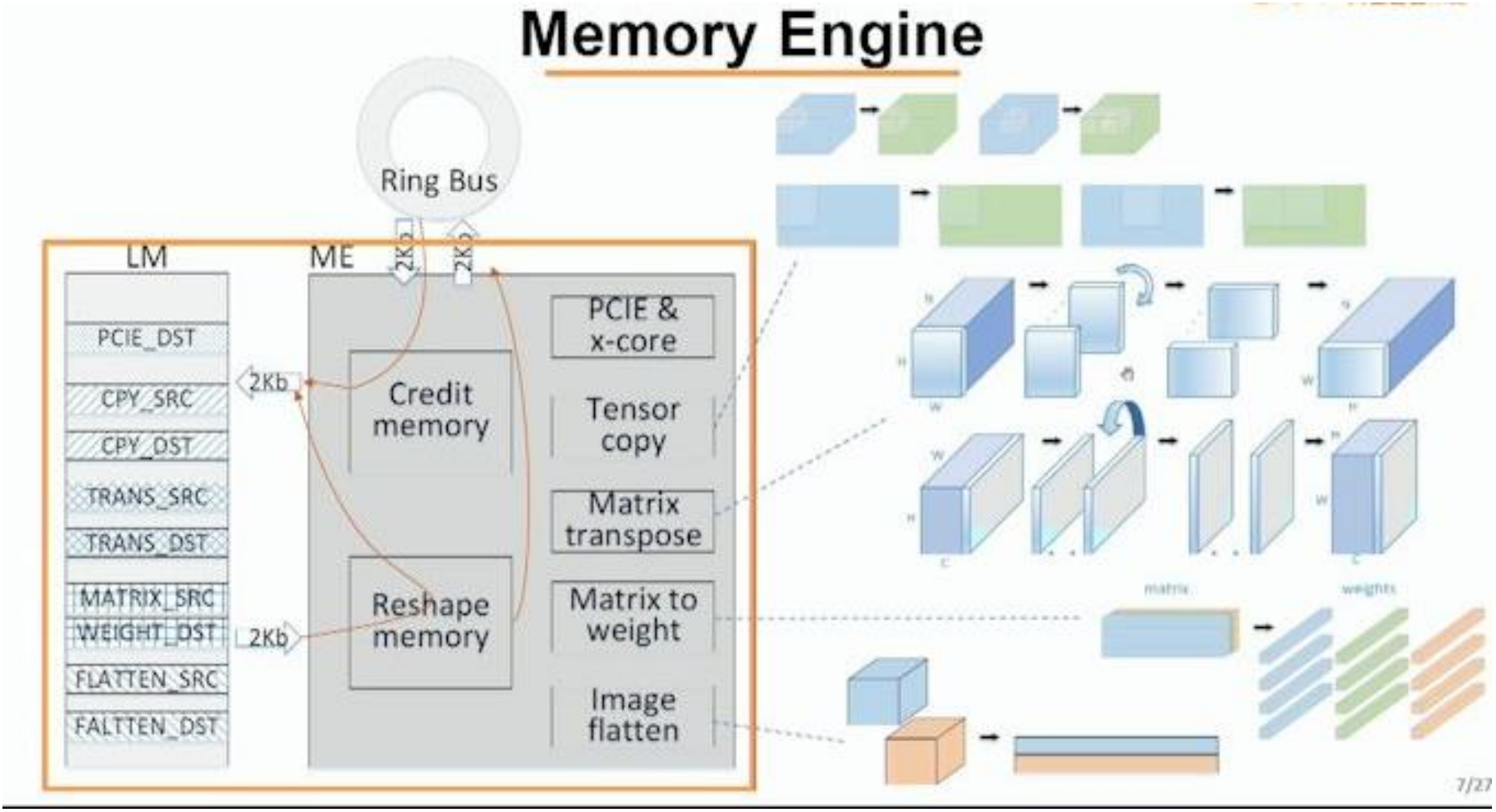
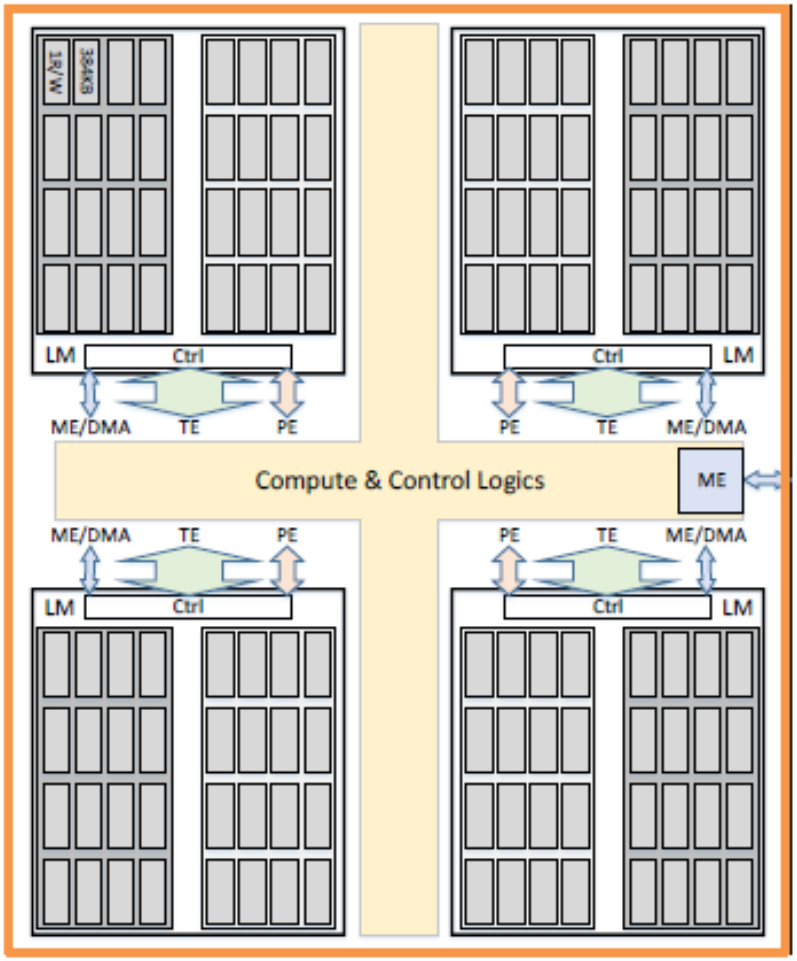


Memory engine can adjust arrangement of data

192MB On-Chip SRAM

- 1-R/W SRAM Modules
  - High density, low power
- 2 blocks for W & A
  - 16 modules in each block
- 4 clusters
  - Distributed shared
- 48MB in each of the 4 cores
  - Memory copy across cores via ME & Ring Bus

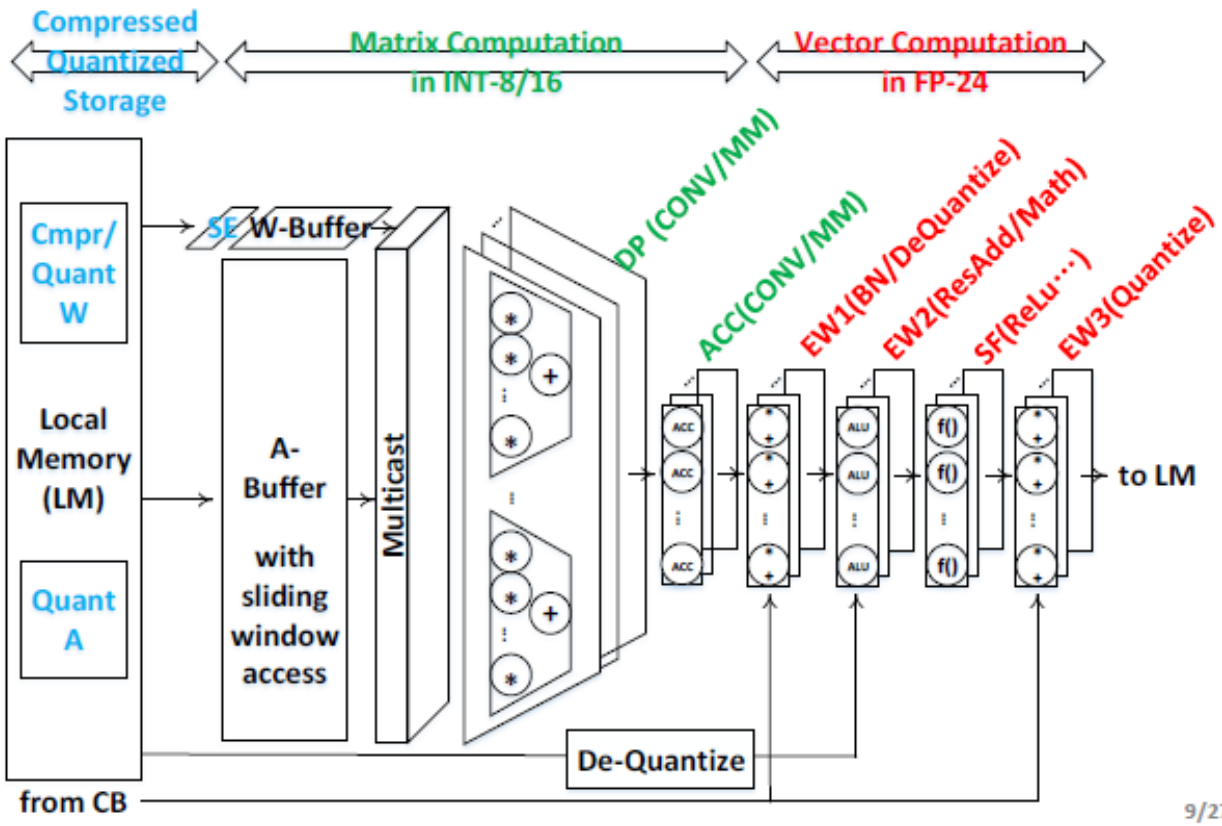


LM Organization in Each Core

384KB*8*16 = 48MB

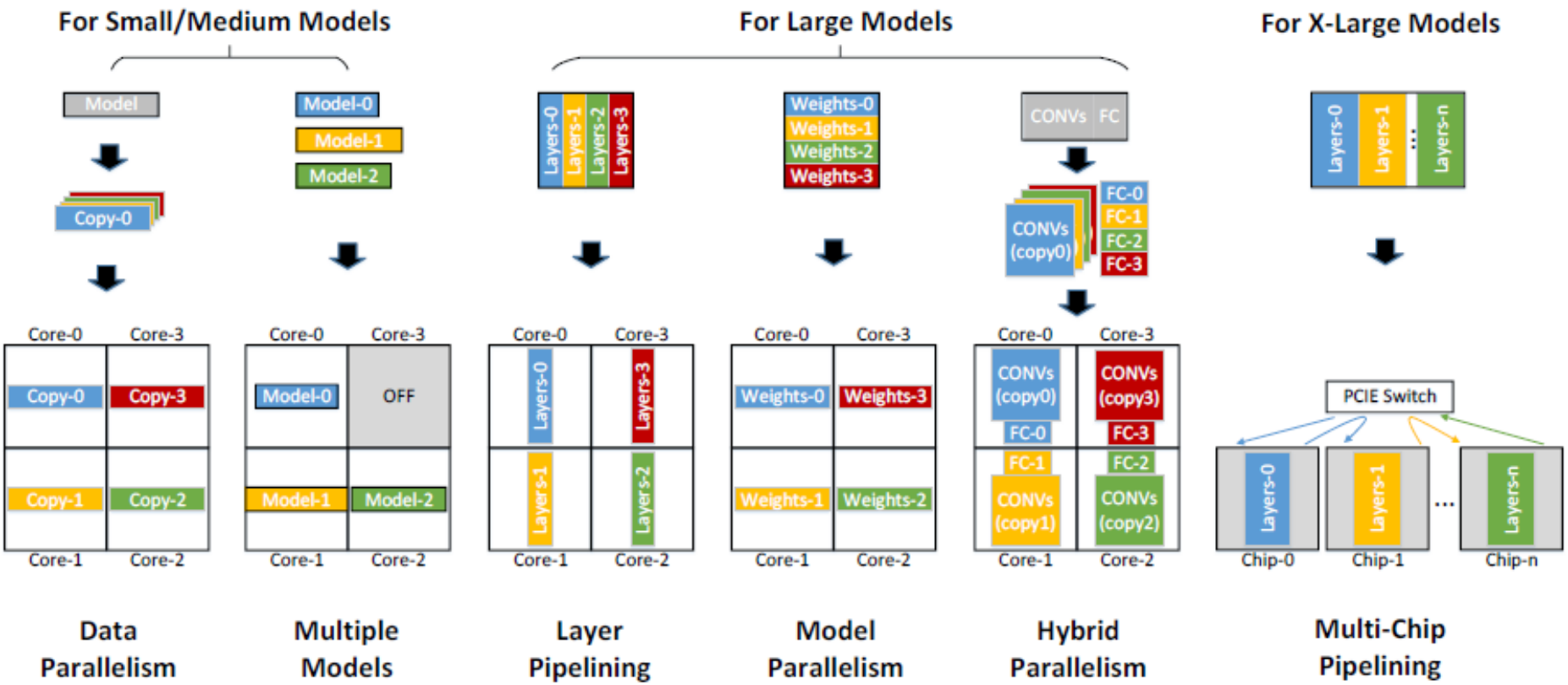
Compressed and Quantized Storage/Processing

- **Compressed** Model
  - Sparsity Engine to unpack with bit-masks
  - Pruning is optional though
- **Quantized** computation and storage
- Vector Unit w/ **FP-24**
  - 1sign.8exp.15man

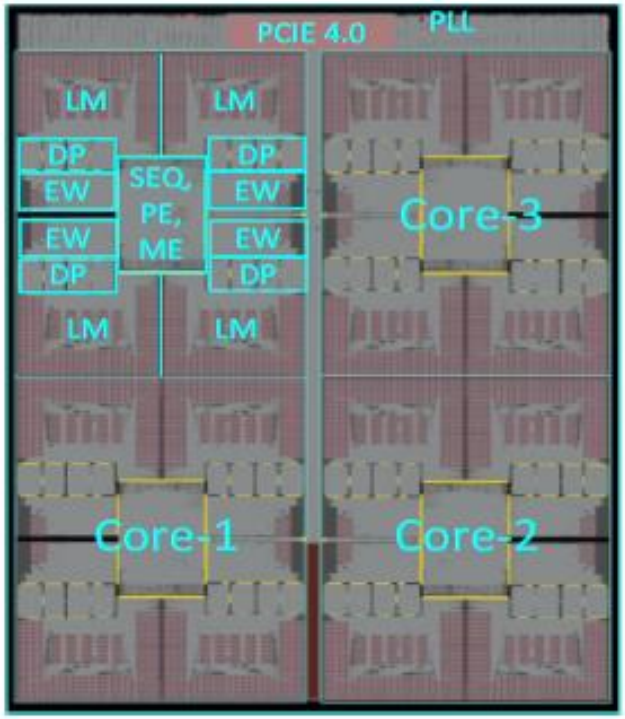


DP = dot-product
Acc = accumulation

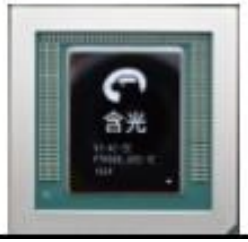
Scalable Task Mapping



Implementation



	NVIDIA T4	NVIDIA V100	Huawei Ascent910	NVIDIA A100	Alibaba Hanguang 800
Peak Performance (INT8 TOPS)	130	125 (FP16)	512	624/1248	825
Frequency (MHz)	1590	1530	1000	1410	700
TDP (Watt)	70	300	350	400	280
Area (mm^2)	545	815	1228	826	709
Process (nm)	TSMC 12nm	TSMC 12nm	TSMC 7+nm	TSMC 7nm	TSMC 12nm



Scalable Performance and Power

- Configurable frequency and voltage
- Can be deployed from data center to large edges

ResNet-50 v1

Configuration	Performance Mode	Efficiency Mode	Low Power Mode 1	Low Power Mode 2	Low Power Mode 3
Frequency (MHz)	700	475	475	475	475
Perf (img/sec)	78,563	53,983	37,500	25,000	12,500
Power (watt)	276	108	75	50	25

Results

Comparative results

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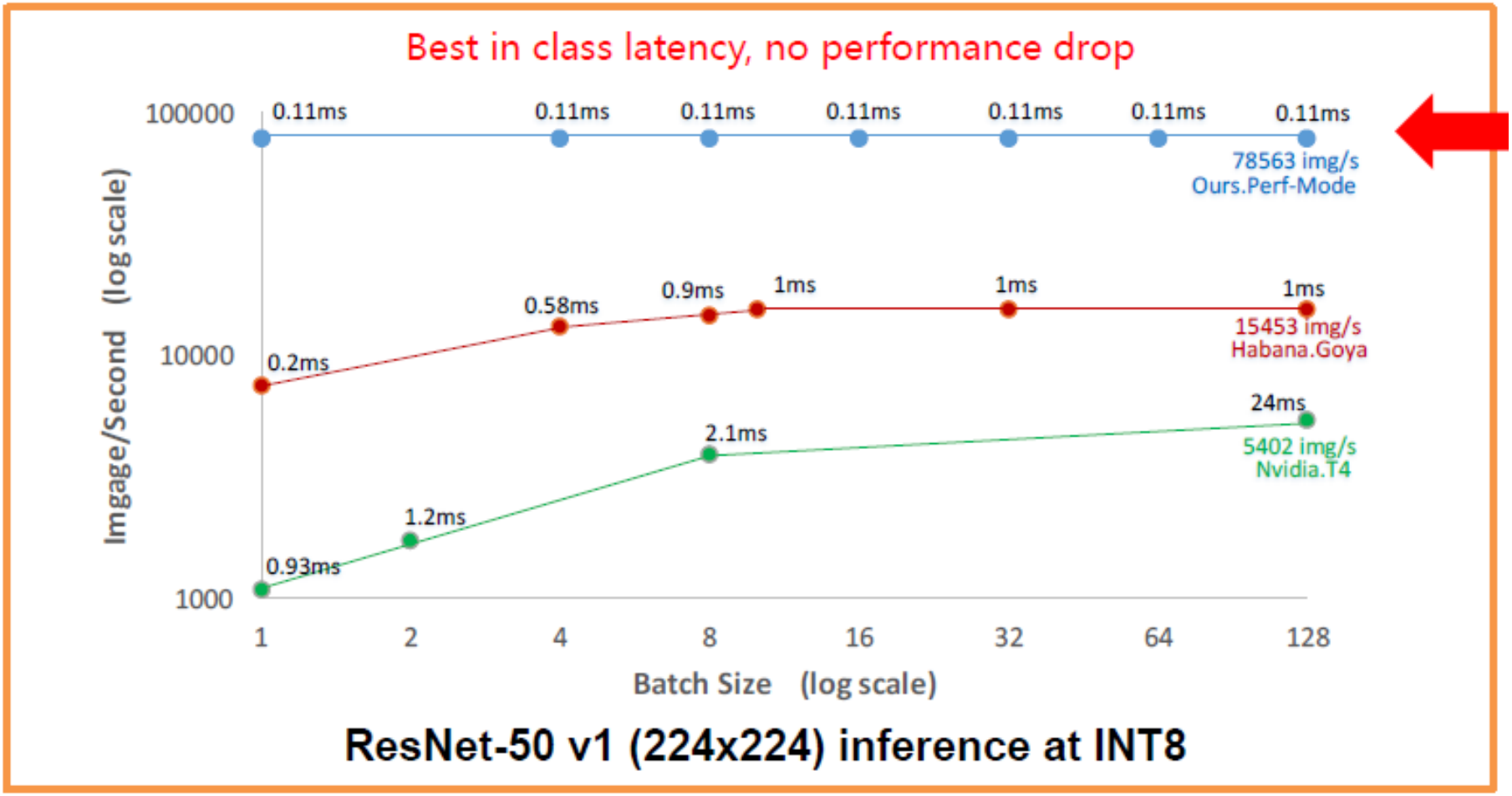
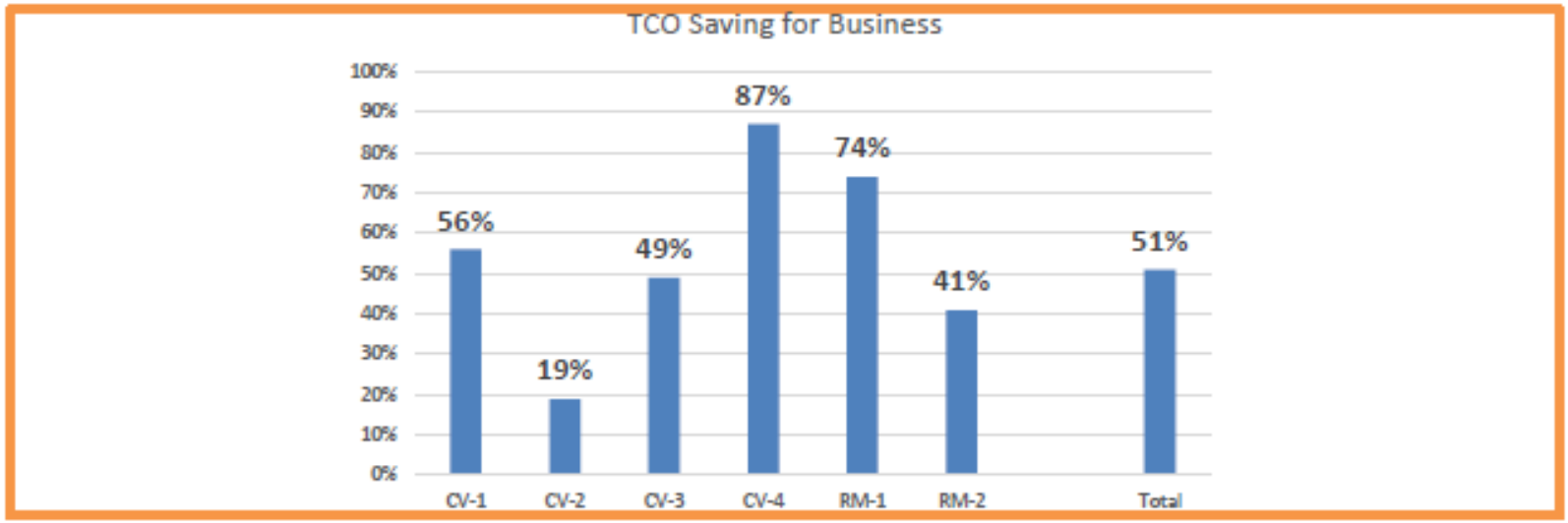
Comparison with GPU

- Using its Turing architecture, Titan RTX has a peak of 261 TOPS, compared to 825 TOPS of HanGuang.
- On ResNet-50, HanGuang achieves its peak performance even at a batch size of one, whereas Titan RTX is much slower for real-time inference and has considerably worse latency.
- GPU requires an expensive high-bandwidth DRAM to compensate for its small on-chip memory, but this arrangement provides more storage for large models.
- HanGuang and Titan RTX have similar die area and power.

Batching-Independent Performance



Deployed Applications via Alibaba Cloud



Top 6 businesses representing 99.1% deployment of Hanguang 800 NPUs.

Business	Major App. Type	GPU Server	NPU Server	Deployment %
CV-1	Image/Video	2x T4	2x 3-core	8.2%
CV-2	Image/Video	8x T4	2x 3-core	28.5%
CV-3	Image/Video	8x T4	2x 3-core	12.5%
CV-4	Image/Video	4x T4	4x 2-core	12.5%
RM-1	Recommender	1x T4	1x 4-core	12.5%
RM-2	Recommender	1x T4	1x 4-core	25.1%

Hangzhou City Brain

- They are using their chip in City Brain project in Hangzhou
- Processes video feeds in real-time for managing traffic

Hanguang 800	GPU
Required chips: 4	Required GPUs: 40
Latency: 150 milliseconds	Latency: 300 milliseconds
Power consumption per route: 1 W	Power consumption per route: 2.8 W

- In pedestrian detection application, 4-core Hanguang 800 can support 100-channel video: 4X better than the mainstream GPU 25-channel inference performance.

References

- https://www.anandtech.com/show/16009/hot-chips-2020-live-blog-alibabas-hanguang-800-npu-500pm-pt
- “Hanguang800 NPU –The Ultimate AI Inference Solution for Data Centers”, Alibaba Group